

Supplementary Material: A formal knowledge base for metagenomics and geochemistry of the Rub' al Khali desert

1 Knowledge-representation patterns

This section specifies the knowledge-representation patterns that are not reproduced in the main article: sampling, protocol execution, taxonomic abundance, XRF geochemistry, the sequencing chain and sequencing quality control, and adds a worked example of the main article's laboratory-control pattern. Each pattern is given as its Description Logic axioms followed by a worked Turtle example taken from the released modules. Every printed triple and every prefixed IRI in this section is resolved against the staged modules by `scripts/validation/verify_manuscript_listings.py`, which fails if a listing drifts from the graph.

Listings use the prefix declarations of Table 1; prepend them to any listing to be executed against a local or deployed graph. The `rak_kb:` prefix covers the knowledge-base namespace without the `RAK_` local-name prefix, which is needed only for the PowerSoil extraction protocol `rak_kb:DNA_ext_powersoil`, the one individual in the released graph minted without it.

1.1 Laboratory-control worked example

Listing 1 instantiates the product-specification and evidence-bearing expected-taxon pattern for the whole-cell D6300 standard. The promoted specification-to-taxon statement supports direct queries, while the reified assertion preserves its subject, relation descriptor, taxon, status and evidence. Manufacturer genomic-DNA percentages remain in `metadata/controls/control_composition.tsv`; the RDF does not reinterpret them as expected 16S read fractions.

Listing 1 Turtle representation of a D6300 composition specification and one expected-taxon assertion.

```
rak:CTL_24320eab3380ef6d405c590160b8d3310c2d641b7621bcef232f6d05cf4c2e60
a rak:0000301 ;
sio:000339
rak:CTL_99a452c18c4a63a400f55b2fe1abbf512ced2a0248d9952c25e441622268d409 .

rak:CTL_99a452c18c4a63a400f55b2fe1abbf512ced2a0248d9952c25e441622268d409
a rak:0000316 ;
sio:000338
rak:CTL_24320eab3380ef6d405c590160b8d3310c2d641b7621bcef232f6d05cf4c2e60 ;
```

Table 1 Prefix declarations used in all Turtle and SPARQL listings.

Prefix	Namespace
rak:	https://rubalkhali.science/kb/RAK_
rak_kb:	https://rubalkhali.science/kb/
sio:	http://semanticscience.org/resource/SIO_
owl:	http://www.w3.org/2002/07/owl#
pato:	http://purl.obolibrary.org/obo/PATO_
uo:	http://purl.obolibrary.org/obo/UO_
envo:	http://purl.obolibrary.org/obo/ENVO_
ncbitaxon:	http://purl.obolibrary.org/obo/NCBITaxon_
obo:	http://purl.obolibrary.org/obo/
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs:	http://www.w3.org/2000/01/rdf-schema#
xsd:	http://www.w3.org/2001/XMLSchema#
geo:	http://www.opengis.net/ont/geosparql#
dcterms:	http://purl.org/dc/terms/

```

dcterms:source
  <https://www.zymoresearch.com/products/zymbiomics-microbial-community-
    standard> ;
rak:2000092 ncbi:taxon:1280 .

rak:CTL_1393f741397200195c65c9357ce358f08ed0b6368bdb5bfd379bf71b03f832bd
  a rak:0000317 ;
rak:2000093
  rak:CTL_99a452c18c4a63a400f55b2fe1abbf512ced2a0248d9952c25e441622268d409 ;
rak:2000094
  rak:CTL_56698f459e82fdb69be8ef81d785e212b8b59d6ec4f88b16dc300b69b1c461c3 ;
rak:2000095 ncbi:taxon:1280 ;
rak:2000090
  rak:CTL_d5aa029f71c6879da7029c52c4aaef883080caf632d3eaa14223a67d0a808d9d ;
sio:000772
  rak:CTL_998abc71455e01cd0175f49f1f32bc70113e47b047b072b30db1f4fe0de5fb13 .

rak:CTL_998abc71455e01cd0175f49f1f32bc70113e47b047b072b30db1f4fe0de5fb13
  sio:000773
  rak:CTL_1393f741397200195c65c9357ce358f08ed0b6368bdb5bfd379bf71b03f832bd .

```

1.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

$$\text{SamplingProcess} \sqsubseteq \text{Process} \quad (1)$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \quad (2)$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget.Site} \sqcap \exists \text{hasOutput.Sample} \quad (3)$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom.Site} \quad (4)$$

Listing 2 is generated from the Trip 1, Site 1 deep-soil record in the development graph.

Listing 2 Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:7000001 a sio:001419 ; # Material sample collection
  sio:000059 rak:6000001 . # hasMember

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000244 rak:1000001 . # isDerivedFrom
```

Read in prose, an instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output. The collection explicitly contains `rak:6000001`, and this deep-soil sample is asserted to be derived from the same site. The redundancy between `hasTarget` on the process and `isDerivedFrom` on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

1.3 Experimental protocol pattern

We model experimental protocols as processes specified by a protocol entity. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO’s general *Process–hasInput/hasOutput–Object* idiom and adds a specification link to a reusable protocol description.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \quad (5)$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \quad (6)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy}.\text{Protocol} \quad (7)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput}.\text{MaterialEntity} \sqcap \exists \text{hasOutput}.\text{MaterialEntity} \quad (8)$$

Listing 3 demonstrates the formalization of a DNA extraction process.

Listing 3 Turtle representation of a DNA extraction protocol.

```
rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:5900001 . # hasAttribute (Concentration Quality)
```

In words, the protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak_kb:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract

carries a concentration quality (`rak:5900001`, class `rak:0000043`) which is itself quantified through a separate measurement process following the pattern of the measurement pattern of the main article. The measurement value that quantifies it is the separate individual `rak:4900001` (class `rak:0000044`), which points to the quality with `sio:000215` and back from it with `sio:000216`. Value and quality are deliberately distinct individuals in every module; conflating them is the single most common way to break the traversal described in the Data Records section of the main article.

1.4 Taxonomic abundance pattern

Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. Here, we use the conservative mapping (Methods, taxonomy normalization) to assign either a validated NCBI class, an unambiguous retained project class, or a deterministic lineage-scoped contextual class as the target of each observation. Unreviewed GTDB candidates are not asserted. Listing 4 shows how an aggregated taxon–rank observation from one profile is represented in the knowledge graph.

For instance, the sequence analysis for a specific profile yields an absolute sequence count associated with an audited taxonomic class. We construct an abundance quality that is an attribute of both that taxon and the source sequence dataset, binding the count back to the biological workflow. Formally, the pattern instantiates SIO’s *Collection–hasMember–Thing* and reciprocal *MeasurementValue–isMeasurementValueOf–Quality* idioms:

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \quad (9)$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \quad (10)$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \quad (11)$$

$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \quad (12)$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AbundanceQuality} \quad (13)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \quad (14)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{hasMeasurementValue.AbundanceObservation} \quad (15)$$

Listing 4 Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000339 rak:L000012 ; # 16S amplicon processing protocol
  sio:000230 rak:7791009 ; # hasInput (FASTQ Dataset, ERR16783308)
  sio:000229 rak:7740001 . # hasOutput (Taxonomic Dataset)

rak:7740001 a rak:0000074 ; # taxon absolute abundance dataset
  sio:000059 rak:V00000001 . # hasMember (Observation Value)

rak:V00000001 a rak:0000076 ; # Absolute abundance value
  sio:000215 rak:Q00000001 ; # isMeasurementValueOf
  rak:2000025 "Domain: Bacteria" ; # classifier lineage
  rak:2000026 "853814.0"^^xsd:double . # absolute read count
```

```

rak:Q00000001 a rak:0000078, pato:0000070 ; # Absolute Abundance Quality
sio:000011 ncbitaxon:2, rak:7791009 ; # taxon and FASTQ bearers
sio:000216 rak:V00000001 . # hasMeasurementValue

```

Verbalized, the bioinformatic-workflow instance `rak:P290001` consumes the FASTQ dataset of run `ERR16783308` (`rak:7791009`, profile `V10Dr1`) and produces the absolute taxonomic dataset `rak:7740001`; that dataset has as one of its members the measurement value `rak:V00000001`, which records an absolute read count of 853,814 for the domain-rank record `Domain: Bacteria`. Its reciprocal quality `rak:Q00000001` is an attribute of both *Bacteria* (`ncbitaxon:2`) and the source FASTQ dataset. Each of the fourteen datasets produced by one workflow instance holds a single rank at one of the two abundance bases, so a rank-restricted query never has to filter observations by lineage depth. One aggregated taxon–rank entry is therefore expanded into a four-node sub-graph that preserves both the classifier lineage and its sequence-dataset provenance. Contextual lineages use the same pattern with their audited project IRI.

1.5 XRF geochemistry pattern

We model every XRF session as a process with multiple concentration-value outputs and a workflow-specific protocol. A laboratory session has an archived soil specimen as input. The field instrument log, by contrast, identifies a test and site but not a physical specimen; the regenerated graph therefore links the session to the recorded sampling site and does not fabricate a collected specimen input. Field-XRF and laboratory-XRF records remain distinguishable even when they report the same analyte. Analyte cross-references to ChEBI [1] and PubChem [2] are maintained in a single mapping ledger. The fail-closed audit resolves every non-null identifier against pinned ChEBI v247 and a dated PubChem snapshot, checking identifier existence, label, formula, charge and atom-versus-oxide entity type. Of 93 instrument channels, 79 have a supported ChEBI cross-reference and 91 have a formula-matching PubChem cross-reference. A missing external identifier remains explicitly null rather than being filled by label similarity. These are `rdfs:seeAlso` annotations for the reporting channel, not assertions that a particular chemical species or mineral occurs in a sample. In particular, the instrument’s P_2O_5 channel is an oxide-equivalent reporting basis; its ChEBI cross-reference has the same empirical ratio (P_4O_{10}) and is not a molecular-presence claim. The instrument’s Light Elements (LE) output is an instrument pseudo-analyte or mixture, not an individual chemical entity, and is therefore not mapped to ChEBI or PubChem as a compound.

The shared output pattern specializes the measurement pattern (the measurement pattern of the main article); only the laboratory specialization requires a soil-sample input:

$$\text{XRFProcess} \sqsubseteq \text{MeasurementProcess} \sqcap \exists \text{hasOutput.Concentration} \quad (16)$$

$$\text{LabXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasInput.SoilSample} \quad (17)$$

$$\text{FieldXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasTarget.SamplingSite} \quad (18)$$

$$\text{Concentration} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf}.\text{AnalyteQuality} \quad (19)$$

Listing 5 is generated from the aluminium output of laboratory-XRF process `rak:P300072` in the current module.

Listing 5 Turtle representation of an XRF geochemistry measurement.

```
rak:P300072 a rak:0000025 ; # XRF Measuring Process
sio:000230 rak:6001732 ; # hasInput (sample V1Sr1)
sio:000339 rak:L_Surface_Lab ; # laboratory protocol
sio:000229 rak:4302121 . # hasOutput

rak:4302121 a rak:0000507 ; # Al concentration value
sio:000215 rak:5302121 ; # isMeasurementValueOf
rak:2000012 "0.78"^^xsd:double . # Concentration

rak:5302121 a rak:0000107 ; # Al concentration quality
sio:000011 rak:6001732 ; # isAttributeOf
sio:000216 rak:4302121 . # hasMeasurementValue
```

In words, this laboratory-XRF process consumes archived specimen `rak:6001732` (sample `V1Sr1`) and produces an aluminium concentration value. The value points to its aluminium quality using `sio:000215`; the quality points back using `sio:000216`. A dedicated scan of every XRF source artifact (the processed laboratory table, the field log and all raw instrument exports) (`scripts/xrf/audit_xrf_unit_evidence.py`) finds no unit declaration attached to any concentration column; the only unit token in the exports is PPM, and it belongs to the lower-limit-of-detection column. The generated module therefore carries no unit assertion for XRF concentrations. Readers should treat them as unitless instrument readings on a consistent scale within a workflow and must not use them for absolute cross-workflow comparisons. In the regenerated module, field sessions use `sio:000291` (*has target*) to identify the recorded site and retain the instrument test identifier; they have no constructed “field deep soil material” specimen input. This distinction also prevents a shared graph pattern from implying numerical interchangeability between workflows.

1.6 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process–hasInput/hasOutput–Object* steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern of the protocol pattern above.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \quad (20)$$

$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput}.\text{SoilSample} \sqcap \exists \text{hasOutput}.\text{DNAExtract} \quad (21)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \quad (22)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput}.\text{DNAExtract} \sqcap \exists \text{hasOutput}.\text{AmpliconLibrary} \quad (23)$$

$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \quad (24)$$

$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput.AmpliconLibrary} \sqcap \exists \text{hasOutput.FASTQDataset} \quad (25)$$

$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \quad (26)$$

$$\text{FASTQDataset} \sqsubseteq \forall \text{hasMember.SequenceRead} \quad (27)$$

Listing 6 instantiates the complete sequencing chain.

Listing 6 Turtle representation of the sequencing workflow from soil sample T1Dr1 to ENA run ERR16061083.

```
rak:P200923 a sio:000994 ; # DNA Extraction Process
sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
sio:000230 rak:6000577 ; # hasInput (T1Dr1 Soil Sample)
sio:000229 rak:6800540 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000065 ; # Library Preparation Process
sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
sio:000230 rak:6800540 ; # hasInput (DNA Extract)
sio:000229 rak:6690001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000066 ; # Sequencing Process
sio:000230 rak:6690001 ; # hasInput (Amplicon Library)
sio:000229 rak:7790001 . # hasOutput (FASTQ Dataset)

rak:7790001 a rak:0000063 ; # FASTQ Dataset
rdfs:seeAlso <https://identifiers.org/insdc.run/ERR16061083> .
```

Read as a chain, the DNA extraction process `rak:P200923` consumes the soil sample `rak:6000577` and produces the DNA extract `rak:6800540`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6690001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7790001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput/hasOutput` pattern, so a single SPARQL traversal can recover the recorded provenance from any node in this chain.

1.7 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the generic measurement pattern of the measurement pattern of the main article. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \quad (28)$$

$$\text{SequencingQCProcess} \sqsubseteq \exists \text{hasInput.FASTQDataset} \sqcap \exists \text{hasOutput.QCMeasurementValue} \quad (29)$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.QCQuality} \quad (30)$$

$$\text{QCQuality} \sqsubseteq \exists \text{isAttributeOf.FASTQDataset} \quad (31)$$

Listing 7 shows the corresponding quality-control pattern.

Listing 7 Turtle representation of forward-read GC content for ENA run ERR16062319.

```
rak:P350001 a rak:0000200 ; # Sequencing QC Process
  sio:000230 rak:7790263 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4450001 . # hasOutput (QC Value)

rak:4450001 a rak:0000217 ; # Forward GC Content Value
  sio:000215 rak:5550001 ; # isMeasurementValueOf
  sio:000221 uo:0000187 ; # Percent
  rak:2000074 "58.0"^^xsd:double . # has GC content

rak:5550001 a rak:0000205 ; # Forward GC Content Quality
  sio:000011 rak:7790263 ; # isAttributeOf (FASTQ Dataset)
  sio:000216 rak:4450001 . # hasMeasurementValue
```

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7790263` as input and produces the measurement value `rak:4450001`, which records a GC content of 58.0 percent and is the measurement value of a forward-read GC-content quality `rak:5550001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

2 Supplementary Tables

Table 2 reports the curated field observations, and Table 3 reports the audited XRF channel cross-references.

Table 2: Site-specific field environmental observations across all expeditions. Values are generated from the immutable source sheets and the versioned correction ledger; a dash denotes an unrecorded or quarantined value. Altitude was not recorded in these field sheets.

Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 1 (2023)	1	17/03/2023	07:58	19.1498 N, 45.1665 E	20.7	928	62.5
Trip 1 (2023)	2	17/03/2023	08:21	19.2035 N, 45.2448 E	21.3	928	60.4
Trip 1 (2023)	3	17/03/2023	08:50	19.3011 N, 45.3844 E	22.6	928	55.4
Trip 1 (2023)	4	17/03/2023	09:20	19.3796 N, 45.5518 E	24	928	51.0
Trip 1 (2023)	5	17/03/2023	10:29	19.4072 N, 45.6595 E	28.8	928	40.6
Trip 1 (2023)	6	17/03/2023	11:07	19.4057 N, 45.8161 E	29	928	42.9
Trip 1 (2023)	7	17/03/2023	11:35	19.4070 N, 45.9539 E	28.7	928	49.4
Trip 1 (2023)	8	17/03/2023	12:05	19.3932 N, 46.1107 E	30.6	928	41.5
Trip 1 (2023)	9	17/03/2023	12:44	19.4112 N, 46.3027 E	35.9	928	36.3
Trip 1 (2023)	10	17/03/2023	13:12	19.4205 N, 46.4608 E	32.5	928	39.0
Trip 1 (2023)	11	17/03/2023	13:39	19.4444 N, 46.6356 E	31.1	930	37.5
Trip 1 (2023)	12	17/03/2023	14:04	19.4779 N, 46.8344 E	34.6	932	30.0
Trip 1 (2023)	13	17/03/2023	15:42	19.5293 N, 47.0533 E	28.9	936	29.2
Trip 1 (2023)	14	17/03/2023	16:09	19.5813 N, 47.2920 E	30.2	938	37.8
Trip 1 (2023)	15	17/03/2023	16:33	19.6137 N, 47.4759 E	31.6	946	33.0
Trip 1 (2023)	16	17/03/2023	16:55	19.6445 N, 47.6261 E	20.7	948	33.3
Trip 1 (2023)	17	17/03/2023	17:23	19.6765 N, 47.8273 E	20.5	952	32.7
Trip 1 (2023)	18	18/03/2023	06:49	19.7325 N, 48.0469 E	20	954	50.0
Trip 1 (2023)	19	18/03/2023	07:15	19.8123 N, 48.2405 E	22.9	960	51.4
Trip 1 (2023)	20	18/03/2023	07:55	19.8693 N, 48.4246 E	24.3	964	39.6
Trip 1 (2023)	21	18/03/2023	08:20	19.9011 N, 48.4865 E	29.4	962	30.9
Trip 1 (2023)	22	18/03/2023	08:45	19.9787 N, 48.6394 E	29	—	25.9
Trip 1 (2023)	23	18/03/2023	09:25	20.0493 N, 48.7733 E	26.8	966	25.2
Trip 1 (2023)	24	18/03/2023	09:48	20.0927 N, 48.8994 E	29.5	966	25.9
Trip 1 (2023)	25	18/03/2023	10:15	20.1003 N, 48.9542 E	27.8	968	27.6
Trip 1 (2023)	26	18/03/2023	10:40	20.1488 N, 49.0787 E	31.7	968	23.1
Trip 1 (2023)	27	18/03/2023	11:18	20.2039 N, 49.1960 E	31.5	970	23.8
Trip 1 (2023)	28	18/03/2023	11:47	20.2438 N, 49.3709 E	31.9	970	24.9
Trip 1 (2023)	29	18/03/2023	12:15	20.2598 N, 49.5675 E	31.7	968	25.7
Trip 1 (2023)	30	18/03/2023	14:39	20.2693 N, 49.6837 E	34.1	964	22.4
Trip 1 (2023)	31	18/03/2023	15:11	20.2946 N, 49.8914 E	33.8	964	22.3

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Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 1 (2023)	32	18/03/2023	15:44	20.3107 N, 50.0930 E	34	968	22.0
Trip 1 (2023)	33	18/03/2023	16:15	20.3175 N, 50.3031 E	34.8	968	20.1
Trip 1 (2023)	34	18/03/2023	16:46	20.3444 N, 50.5154 E	34.1	970	21.1
Trip 1 (2023)	35	18/03/2023	17:24	20.3718 N, 50.7300 E	33.7	976	20.8
Trip 1 (2023)	36	19/03/2023	09:40	20.4232 N, 51.1448 E	29.8	982	15.9
Trip 1 (2023)	37	19/03/2023	10:10	20.4542 N, 51.3449 E	33.5	984	14.4
Trip 1 (2023)	38	19/03/2023	10:30	20.4610 N, 51.3702 E	31.9	984	15.1
Trip 1 (2023)	39	19/03/2023	11:05	20.5102 N, 51.5974 E	36.4	984	16.0
Trip 1 (2023)	40	19/03/2023	11:40	20.5472 N, 51.8256 E	30.5	980	30.4
Trip 1 (2023)	41	19/03/2023	12:16	20.5398 N, 51.9461 E	35.4	980	19.8
Trip 1 (2023)	42	19/03/2023	12:41	20.5552 N, 52.1240 E	35.4	984	20.5
Trip 1 (2023)	43	19/03/2023	13:05	20.5676 N, 52.2241 E	35.2	984	22.8
Trip 1 (2023)	44	19/03/2023	13:40	20.5678 N, 52.4104 E	38.4	984	19.6
Trip 1 (2023)	45	19/03/2023	13:59	20.5635 N, 52.4678 E	37.1	986	20.1
Trip 1 (2023)	46	19/03/2023	14:29	20.5537 N, 52.6692 E	35.7	987	21.1
Trip 1 (2023)	47	19/03/2023	15:00	20.5418 N, 52.8688 E	37.3	986	20.3
Trip 1 (2023)	48	19/03/2023	17:00	20.5686 N, 53.0940 E	35.3	984	23.7
Trip 1 (2023)	49	19/03/2023	17:30	20.6447 N, 53.2879 E	35	986	21.7
Trip 1 (2023)	50	20/03/2023	07:14	20.7139 N, 53.4123 E	29.8	988	31.3
Trip 1 (2023)	51	20/03/2023	07:37	20.7987 N, 53.4870 E	30.2	992	32.5
Trip 1 (2023)	52	20/03/2023	08:02	20.8515 N, 53.7579 E	31.1	992	29.7
Trip 1 (2023)	53	20/03/2023	09:03	20.8515 N, 53.7579 E	34.8	992	26.0
Trip 1 (2023)	54	20/03/2023	10:03	20.9202 N, 53.9622 E	32.1	992	26.7
Trip 1 (2023)	55	20/03/2023	10:24	20.9023 N, 54.0024 E	33.2	990	27.9
Trip 1 (2023)	56	20/03/2023	11:05	20.8216 N, 54.2239 E	30	988	35.5
Trip 1 (2023)	57	20/03/2023	11:28	20.7350 N, 54.4176 E	31.3	988	38.1
Trip 1 (2023)	58	20/03/2023	11:50	20.6654 N, 54.5933 E	33.4	986	39.4
Trip 1 (2023)	59	20/03/2023	12:04	20.6563 N, 54.6359 E	37.1	986	43.1
Trip 1 (2023)	60	20/03/2023	12:49	20.7064 N, 54.7321 E	37.1	986	34.5
Trip 1 (2023)	water well	20/03/2023	13:16	20.7159 N, 54.7997 E	35.6	984	31.4
Trip 1 (2023)	road	20/03/2023	13:53	20.8258 N, 54.9889 E	35.1	982	32.2
Trip 1 (2023)	road	20/03/2023	14:20	20.9341 N, 55.1261 E	36	982	26.3
Trip 1 (2023)	camground	20/03/2023	14:45	21.0311 N, 55.1598 E	37	984	26.0
Trip 2 (2023)	1	14/07/2023	06:10	19.1498 N, 45.1665 E	34.5	—	—
Trip 2 (2023)	2	14/07/2023	07:12	19.2035 N, 45.2448 E	36.7	—	—

Continued on next page

Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 2 (2023)	3	14/07/2023	07:16	19.3011 N, 45.3844 E	37.0	—	—
Trip 2 (2023)	4	14/07/2023	07:36	19.3796 N, 45.5518 E	38.6	—	—
Trip 2 (2023)	5	14/07/2023	07:43	19.4072 N, 45.6595 E	39.2	—	—
Trip 2 (2023)	6	14/07/2023	08:58	19.4057 N, 45.8161 E	41.2	—	—
Trip 2 (2023)	7	14/07/2023	09:20	19.4070 N, 45.9539 E	41.3	—	—
Trip 2 (2023)	8	14/07/2023	10:20	19.3932 N, 46.1107 E	41.9	—	—
Trip 3 (2024)	1	17/02/2024	06:54	19.1498 N, 45.1665 E	21.9	898	55
Trip 3 (2024)	2	17/02/2024	07:21	19.2035 N, 45.2448 E	26.6	904	40.9
Trip 3 (2024)	3	17/02/2024	08:05	19.3011 N, 45.3844 E	28.3	914	38.5
Trip 3 (2024)	4	17/02/2024	08:40	19.3796 N, 45.5518 E	30.2	922	37.5
Trip 3 (2024)	5	17/02/2024	09:51	19.4072 N, 45.6595 E	35.2	926	19.5
Trip 3 (2024)	6	17/02/2024	10:45	19.4057 N, 45.8161 E	37.1	942	17.6
Trip 3 (2024)	7	17/02/2024	11:28	19.4070 N, 45.9539 E	37	926	19.6
Trip 3 (2024)	8	17/02/2024	11:50	19.3932 N, 46.1107 E	39	922	19
Trip 3 (2024)	9	17/02/2024	12:31	19.4112 N, 46.3027 E	36.4	924	21.5
Trip 3 (2024)	10	17/02/2024	13:22	19.4205 N, 46.4608 E	37.1	926	18
Trip 3 (2024)	11	17/02/2024	13:46	19.4444 N, 46.6356 E	41.2	930	21
Trip 3 (2024)	12	17/02/2024	14:04	19.4779 N, 46.8344 E	39.9	932	15
Trip 3 (2024)	13	17/02/2024	15:20	19.5293 N, 47.0533 E	40	940	13.5
Trip 3 (2024)	14	17/02/2024	15:50	19.5813 N, 47.2920 E	41.2	940	14.9
Trip 3 (2024)	15	17/02/2024	16:13	19.6137 N, 47.4759 E	38.8	944	13.5
Trip 3 (2024)	16	17/02/2024	16:34	19.6445 N, 47.6261 E	39.4	948	11.9
Trip 3 (2024)	17	18/02/2024	08:42	19.6765 N, 47.8273 E	23	956	27
Trip 3 (2024)	18	18/02/2024	09:54	19.7325 N, 48.0469 E	29.2	960	18.7
Trip 3 (2024)	19	18/02/2024	10:25	19.8123 N, 48.2405 E	27.4	962	21.2
Trip 3 (2024)	20	18/02/2024	11:05	19.8693 N, 48.4246 E	32.9	—	17.7
Trip 3 (2024)	21	18/02/2024	12:08	19.9011 N, 48.4865 E	31.3	964	31.321
Trip 3 (2024)	22	18/02/2024	12:52	19.9787 N, 48.6394 E	31.3	968	21.3
Trip 3 (2024)	23	18/02/2024	13:35	20.0493 N, 48.7733 E	24.4	966	30.3
Trip 3 (2024)	24	18/02/2024	14:18	20.0927 N, 48.8994 E	25.5	966	31.3
Trip 3 (2024)	25	18/02/2024	14:38	20.1003 N, 48.9542 E	31	966	24.3
Trip 3 (2024)	26	18/02/2024	15:12	20.1488 N, 49.0787 E	31.9	966	25.6
Trip 3 (2024)	27	18/02/2024	15:45	20.2039 N, 49.1960 E	32.4	968	23.7
Trip 3 (2024)	28	18/02/2024	16:24	20.2438 N, 49.3709 E	27.4	968	27.8
Trip 3 (2024)	29	18/02/2024	16:56	20.2598 N, 49.5675 E	27.3	968	27

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Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 3 (2024)	30	19/02/2024	08:10	20.2693 N, 49.6837 E	36.8	976	20.2
Trip 3 (2024)	31	19/02/2024	08:43	20.2946 N, 49.8914 E	23.1	978	29.3
Trip 3 (2024)	32	19/02/2024	09:20	20.3107 N, 50.0930 E	26.7	982	23
Trip 3 (2024)	33	19/02/2024	10:07	20.3175 N, 50.3031 E	26.6	984	20.9
Trip 3 (2024)	34	19/02/2024	10:52	20.3444 N, 50.5154 E	29.9	982	17.7
Trip 3 (2024)	35	19/02/2024	11:30	20.3718 N, 50.7300 E	28.7	984	17.9
Trip 3 (2024)	36	19/02/2024	13:58	20.4232 N, 51.1448 E	28.2	982	16.1
Trip 3 (2024)	37	19/02/2024	14:52	20.4542 N, 51.3449 E	29.1	984	15.5
Trip 3 (2024)	38	19/02/2024	15:01	20.4610 N, 51.3702 E	29.3	984	15.1
Trip 3 (2024)	39	19/02/2024	15:56	20.5102 N, 51.5974 E	28.8	984	16
Trip 3 (2024)	40	19/02/2024	16:42	20.5472 N, 51.8256 E	28.2	986	17.3
Trip 3 (2024)	41	19/02/2024	17:14	20.5398 N, 51.9461 E	29.4	988	16.2
Trip 3 (2024)	42	20/02/2024	08:15	20.5552 N, 52.1240 E	24.6	990	34.2
Trip 3 (2024)	43	20/02/2024	08:38	20.5676 N, 52.2241 E	27	992	27.8
Trip 3 (2024)	44	20/02/2024	09:22	20.5678 N, 52.4104 E	29.1	994	22
Trip 3 (2024)	45	20/02/2024	09:44	20.5635 N, 52.4678 E	32.4	996	18.3
Trip 3 (2024)	46	20/02/2024	10:44	20.5537 N, 52.6692 E	32.3	994	17.4
Trip 3 (2024)	47	20/02/2024	11:24	20.5418 N, 52.8688 E	33.2	994	17.2
Trip 3 (2024)	48	20/02/2024	12:54	20.5686 N, 53.0940 E	49.4	994	9.5
Trip 3 (2024)	49	20/02/2024	14:21	20.6447 N, 53.2879 E	34	994	13.4
Trip 3 (2024)	50	20/02/2024	14:51	20.7139 N, 53.4123 E	31.5	996	21.6
Trip 3 (2024)	51	20/02/2024	16:11	20.7987 N, 53.4870 E	35.9	994	16.5
Trip 3 (2024)	52	20/02/2024	06:57	20.8515 N, 53.7579 E	34.3	994	17.5
Trip 3 (2024)	53	21/02/2024	06:06	20.8515 N, 53.7579 E	21.9	990	35
Trip 3 (2024)	54	21/02/2024	15:58	20.9202 N, 53.9622 E	32.2	998	23.5
Trip 3 (2024)	55	21/02/2024	15:41	20.9023 N, 54.0024 E	31.3	998	14.3
Trip 3 (2024)	56	21/02/2024	15:10	20.8216 N, 54.2239 E	31.6	998	13.5
Trip 3 (2024)	57	21/02/2024	14:48	20.7350 N, 54.4176 E	34.2	996	16.3
Trip 3 (2024)	58	21/02/2024	14:29	20.6654 N, 54.5933 E	35.6	992	16.4
Trip 3 (2024)	59	21/02/2024	14:15	20.6563 N, 54.6359 E	36.8	992	14.3
Trip 3 (2024)	60	21/02/2024	13:20	20.7064 N, 54.7321 E	37.1	992	14.3
Trip 3 (2024)	oill spill	17/02/2024	10:25	19.4054 N, 45.7271 E	35.1	—	20
Trip 3 (2024)	water well	17/02/2024	14:23	19.5014 N, 46.9015 E	39.2	—	15.9
Trip 3 (2024)	Camell1	17/02/2024	16:49	19.6540 N, 47.6883 E	37.6	—	25
Trip 3 (2024)	plant gradient	18/02/2024	09:25	19.7092 N, 47.9593 E	25	958	24

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Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 3 (2024)	hot spring	19/02/2024	12:25	20.3972 N, 50.9655 E	32.1	984	14.8
Trip 1 auxiliary/revisit (2023)	12.5+A	17/03/2023	14:22	19.5015 N, 46.9014 E	33	934	30
Trip 1 auxiliary/revisit (2023)	19.5	18/03/2023	07:20	19.8193 N, 48.2402 E	21.8	960	61.9
Trip 1 auxiliary/revisit (2023)	27+	18/03/2023	11:21	20.2039 N, 49.1960 E	31.5	970	23.8
Trip 1 auxiliary/revisit (2023)	31urine	18/03/2023	15:14	20.2946 N, 49.8914 E	33.8	964	22.3
Trip 1 auxiliary/revisit (2023)	32+	18/03/2023	15:44	20.3107 N, 50.0930 E	34	968	22
Trip 1 auxiliary/revisit (2023)	41+	19/03/2023	12:16	20.5398 N, 51.9461 E	35.4	980	19.8
Trip 1 auxiliary/revisit (2023)	47+	19/03/2023	15:00	20.5418 N, 52.8688 E	37.3	986	20.3
Trip 1 auxiliary/revisit (2023)	47+grass	19/03/2023	15:00	20.5418 N, 52.8688 E	37.3	986	20.3
Trip 1 auxiliary/revisit (2023)	48+	19/03/2023	17:00	20.5686 N, 53.0940 E	35.3	984	23.7
Trip 1 auxiliary/revisit (2023)	48+	19/03/2023	17:00	20.5686 N, 53.0940 E	35.3	984	23.7
Trip 1 auxiliary/revisit (2023)	52AD	20/03/2023	08:02	20.8515 N, 53.7579 E	31.1	992	29.7
Trip 1 auxiliary/revisit (2023)	54F	20/03/2023	10:16	20.9202 N, 53.9622 E	34.8	990	26.9
Trip 1 auxiliary/revisit (2023)	35 + WaterWellRestStop	18/03/2023	18:48	20.3972 N, 50.9657 E	—	976	—
Trip 1 auxiliary/revisit (2023)	35 + hot spring filter1	19/03/2023	08:03	20.3972 N, 50.9657 E	28.3	980	17.7
Trip 1 auxiliary/revisit (2023)	35 + hot spring filter 2	19/03/2023	08:11	20.3972 N, 50.9657 E	28.3	980	17.7
Trip 4 (2024)	1	29/08/2024	06:04	19.1498 N, 45.1665 E	30.83	—	19.4
Trip 4 (2024)	2	29/08/2024	06:24	19.2035 N, 45.2448 E	31.5	—	18.4
Trip 4 (2024)	3	29/08/2024	06:53	19.3011 N, 45.3844 E	33.61	—	16.5
Trip 4 (2024)	4	29/08/2024	07:15	19.3796 N, 45.5518 E	35.11	—	15.5
Trip 4 (2024)	5	29/08/2024	07:29	19.4072 N, 45.6595 E	36.67	—	14
Trip 4 (2024)	6	29/08/2024	08:07	19.4057 N, 45.8161 E	37.06	—	13.3
Trip 4 (2024)	7	29/08/2024	08:32	19.4070 N, 45.9539 E	37.56	—	13
Trip 4 (2024)	8	29/08/2024	08:57	19.3932 N, 46.1107 E	37.39	—	14.2
Trip 4 (2024)	9	29/08/2024	09:28	19.4112 N, 46.3027 E	39	—	13.5
Trip 4 (2024)	10	29/08/2024	09:50	19.4205 N, 46.4608 E	39.56	—	13.4
Trip 4 (2024)	11	29/08/2024	10:14	19.4444 N, 46.6356 E	42	—	12.4
Trip 4 (2024)	12	29/08/2024	10:35	19.4779 N, 46.8344 E	42.94	—	21.5
Trip 4 (2024)	13	29/08/2024	11:30	19.5293 N, 47.0533 E	41.94	—	16.5
Trip 4 (2024)	14	29/08/2024	11:50	19.5813 N, 47.2920 E	42.56	—	18
Trip 4 (2024)	15	29/08/2024	12:06	19.6137 N, 47.4759 E	44.11	—	15.2
Trip 4 (2024)	16	29/08/2024	12:21	19.6445 N, 47.6261 E	43.5	—	15.7
Trip 4 (2024)	17	29/08/2024	12:49	19.6765 N, 47.8273 E	43.11	—	15.2
Trip 4 (2024)	18	29/08/2024	13:24	19.7325 N, 48.0469 E	43.83	—	12.8
Trip 4 (2024)	19	29/08/2024	13:52	19.8123 N, 48.2405 E	42.83	—	14.1

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Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 4 (2024)	20	29/08/2024	13:57	19.8693 N, 48.4246 E	41.94	—	14.5
Trip 4 (2024)	21	29/08/2024	14:32	19.9011 N, 48.4865 E	41.72	—	13.1
Trip 4 (2024)	22	29/08/2024	14:50	19.9787 N, 48.6394 E	43.33	—	13
Trip 4 (2024)	23	29/08/2024	15:21	20.0493 N, 48.7733 E	44.28	—	12
Trip 4 (2024)	24	29/08/2024	15:55	20.0927 N, 48.8994 E	43.17	—	12.7
Trip 4 (2024)	25	29/08/2024	16:45	20.1003 N, 48.9542 E	43.5	—	11.6
Trip 4 (2024)	26	29/08/2024	17:02	20.1488 N, 49.0787 E	42.83	—	12.3
Trip 4 (2024)	27	29/08/2024	17:28	20.2039 N, 49.1960 E	41.83	—	12.6
Trip 4 (2024)	28	29/08/2024	17:56	20.2438 N, 49.3709 E	40.83	—	13.3
Trip 4 (2024)	29	29/08/2024	18:21	20.2598 N, 49.5675 E	40.78	—	13.3
Trip 4 (2024)	30	30/08/2024	05:51	20.2693 N, 49.6837 E	36.5	—	27.7
Trip 4 (2024)	31	30/08/2024	06:08	20.2946 N, 49.8914 E	35.89	—	27.3
Trip 4 (2024)	32	30/08/2024	06:30	20.3107 N, 50.0930 E	35.72	—	21.9
Trip 4 (2024)	33	30/08/2024	06:50	20.3175 N, 50.3031 E	35.89	—	23
Trip 4 (2024)	34	30/08/2024	07:11	20.3444 N, 50.5154 E	36.39	—	23.5
Trip 4 (2024)	35	30/08/2024	07:25	20.3718 N, 50.7300 E	38.5	—	21.2
Trip 4 (2024)	36	30/08/2024	08:01	20.4232 N, 51.1448 E	39.28	—	31.8
Trip 4 (2024)	37	30/08/2024	08:47	20.4542 N, 51.3449 E	39.44	—	23.2
Trip 4 (2024)	38	30/08/2024	09:14	20.4610 N, 51.3702 E	41.06	—	20
Trip 4 (2024)	39	30/08/2024	09:20	20.5102 N, 51.5974 E	42.44	—	19.5
Trip 4 (2024)	40	30/08/2024	09:49	20.5472 N, 51.8256 E	42.06	—	19
Trip 4 (2024)	41	30/08/2024	10:25	20.5398 N, 51.9461 E	43.17	—	17
Trip 4 (2024)	42	30/08/2024	10:50	20.5552 N, 52.1240 E	43.67	—	16.2
Trip 4 (2024)	43	30/08/2024	11:19	20.5676 N, 52.2241 E	42.78	—	17
Trip 4 (2024)	44	30/08/2024	11:40	20.5678 N, 52.4104 E	41.94	—	18
Trip 4 (2024)	45	30/08/2024	12:12	20.5635 N, 52.4678 E	40.94	—	17.1
Trip 4 (2024)	46	30/08/2024	12:22	20.5537 N, 52.6692 E	44	—	15.7
Trip 4 (2024)	47	30/08/2024	13:16	20.5418 N, 52.8688 E	42.94	—	15.3
Trip 4 (2024)	48	30/08/2024	13:57	20.5686 N, 53.0940 E	42.5	—	18.2
Trip 4 (2024)	49	30/08/2024	14:28	20.6447 N, 53.2879 E	44	—	14.4
Trip 4 (2024)	50	31/08/2024	06:08	20.7139 N, 53.4123 E	35.61	—	25.1
Trip 4 (2024)	51	31/08/2024	06:34	20.7987 N, 53.4870 E	36.11	—	25
Trip 4 (2024)	52	31/08/2024	06:54	20.8278 N, 53.5784 E	36.28	—	25.5
Trip 4 (2024)	53	31/08/2024	09:00	20.8515 N, 53.7579 E	42.61	—	17.5
Trip 4 (2024)	54	31/08/2024	09:34	20.9202 N, 53.9622 E	42.5	—	17.5

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Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 4 (2024)	55	31/08/2024	09:48	20.9023 N, 54.0024 E	40.72	—	20
Trip 4 (2024)	56	31/08/2024	10:16	20.8216 N, 54.2239 E	40.44	—	21.4
Trip 4 (2024)	57	31/08/2024	10:40	20.7350 N, 54.4176 E	42.78	—	18
Trip 4 (2024)	58	31/08/2024	10:58	20.6654 N, 54.5933 E	43.5	—	15
Trip 4 (2024)	59	31/08/2024	11:06	20.6563 N, 54.6359 E	43	—	14.9
Trip 4 (2024)	60	31/08/2024	11:28	20.7064 N, 54.7321 E	42.89	—	18.1
Trip 5 (2025)	1	12/10/2025	07:32	19.1498 N, 45.1665 E	26.9	894	24.3
Trip 5 (2025)	2	12/10/2025	08:05	19.2035 N, 45.2448 E	27.8	902	23.4
Trip 5 (2025)	3	12/10/2025	08:38	19.3011 N, 45.3844 E	29	912	23
Trip 5 (2025)	4	12/10/2025	09:20	19.3796 N, 45.5518 E	32.4	922	18.8
Trip 5 (2025)	5	12/10/2025	10:00	19.4072 N, 45.6595 E	33.4	924	19.2
Trip 5 (2025)	6	12/10/2025	10:50	19.4057 N, 45.8161 E	36.7	916	14.1
Trip 5 (2025)	7	12/10/2025	11:18	19.4070 N, 45.9539 E	35	918	13.4
Trip 5 (2025)	8	12/10/2025	11:55	19.3932 N, 46.1107 E	34	918	12.9
Trip 5 (2025)	9	12/10/2025	12:33	19.4112 N, 46.3027 E	35.5	918	12.5
Trip 5 (2025)	10	12/10/2025	12:58	19.4205 N, 46.4608 E	35.6	920	10.5
Trip 5 (2025)	11	12/10/2025	13:30	19.4444 N, 46.6356 E	35.6	926	10.7
Trip 5 (2025)	12	12/10/2025	13:57	19.4779 N, 46.8344 E	36.3	928	10.3
Trip 5 (2025)	13	12/10/2025	15:25	19.5293 N, 47.0533 E	36.5	934	10.1
Trip 5 (2025)	14	12/10/2025	15:51	19.5813 N, 47.2920 E	35	936	11.9
Trip 5 (2025)	15	12/10/2025	16:22	19.6137 N, 47.4759 E	34.7	940	11.9
Trip 5 (2025)	16	12/10/2025	16:42	19.6445 N, 47.6261 E	33.9	942	13.2
Trip 5 (2025)	17	13/10/2025	07:05	19.6765 N, 47.8273 E	26.4	948	32.1
Trip 5 (2025)	18	13/10/2025	07:42	19.7325 N, 48.0469 E	26.6	950	27
Trip 5 (2025)	19	13/10/2025	08:15	19.8123 N, 48.2405 E	28.2	954	27.5
Trip 5 (2025)	20	13/10/2025	09:04	19.8693 N, 48.4246 E	32.9	958	23.5
Trip 5 (2025)	21	13/10/2025	09:30	19.9011 N, 48.4865 E	33	958	25
Trip 5 (2025)	22	13/10/2025	10:10	19.9787 N, 48.6394 E	35.5	962	20.8
Trip 5 (2025)	23	13/10/2025	10:56	20.0493 N, 48.7733 E	36.1	960	21.6
Trip 5 (2025)	24	13/10/2025	11:39	20.0927 N, 48.8994 E	36.3	958	18.2
Trip 5 (2025)	25	13/10/2025	11:55	20.1003 N, 48.9542 E	35	960	18.4
Trip 5 (2025)	26	13/10/2025	12:16	20.1488 N, 49.0787 E	34.3	960	19.5
Trip 5 (2025)	27	13/10/2025	12:42	20.2039 N, 49.1960 E	36.4	960	16.6
Trip 5 (2025)	28	13/10/2025	13:08	20.2438 N, 49.3709 E	38.8	960	15.5
Trip 5 (2025)	29	13/10/2025	13:56	20.2598 N, 49.5675 E	37.5	966	14.9

Continued on next page

Expedition	Site	Date	Time	Coordinates	Temp. (°C)	Press. (mbar)	RH (%)
Trip 5 (2025)	30	13/10/2025	14:13	20.2693 N, 49.6837 E	37.5	966	14.2
Trip 5 (2025)	31	13/10/2025	14:46	20.2946 N, 49.8914 E	35.9	968	17.5
Trip 5 (2025)	32	13/10/2025	15:38	20.3107 N, 50.0930 E	39	970	12.8
Trip 5 (2025)	33	13/10/2025	16:10	20.3175 N, 50.3031 E	37.9	972	11.9
Trip 5 (2025)	34	13/10/2025	16:58	20.3444 N, 50.5154 E	34.3	974	17.9
Trip 5 (2025)	35	14/10/2025	06:58	20.3718 N, 50.7300 E	35.1	982	27.1
Trip 5 (2025)	36	14/10/2025	08:36	20.4232 N, 51.1448 E	33.2	982	22.6
Trip 5 (2025)	37	14/10/2025	09:01	20.4542 N, 51.3449 E	32.3	984	23.7
Trip 5 (2025)	38	14/10/2025	09:13	20.4610 N, 51.3702 E	33.2	984	22.3
Trip 5 (2025)	39	14/10/2025	09:50	20.5102 N, 51.5974 E	37.2	986	22.1
Trip 5 (2025)	40	14/10/2025	10:40	20.5472 N, 51.8256 E	39	984	—
Trip 5 (2025)	41	14/10/2025	11:04	20.5398 N, 51.9461 E	37.1	986	20.8
Trip 5 (2025)	42	14/10/2025	11:40	20.5552 N, 52.1240 E	36	984	22.6
Trip 5 (2025)	43	14/10/2025	12:07	20.5676 N, 52.2241 E	40.4	984	19.7
Trip 5 (2025)	44	14/10/2025	13:03	20.5678 N, 52.4104 E	37.1	984	19.3
Trip 5 (2025)	45	14/10/2025	13:16	20.5635 N, 52.4678 E	39.5	984	17.6
Trip 5 (2025)	46	14/10/2025	14:02	20.5537 N, 52.6692 E	38.6	982	17.2
Trip 5 (2025)	47	14/10/2025	14:41	20.5418 N, 52.8688 E	36.2	984	23
Trip 5 (2025)	48	14/10/2025	15:30	20.5686 N, 53.0940 E	36.7	986	19.2
Trip 5 (2025)	49	14/10/2025	16:20	20.6447 N, 53.2879 E	35.2	986	20.4
Trip 5 (2025)	50	14/10/2025	17:00	20.7139 N, 53.4123 E	34.7	984	20.9
Trip 5 (2025)	51	15/10/2025	06:41	20.7987 N, 53.4870 E	31.1	990	36.3
Trip 5 (2025)	52	15/10/2025	07:09	20.8278 N, 53.5784 E	33.5	990	34.1
Trip 5 (2025)	53	15/10/2025	07:59	20.8515 N, 53.7579 E	33.7	990	32.4
Trip 5 (2025)	54	15/10/2025	09:14	20.9202 N, 53.9622 E	37.5	992	24.2
Trip 5 (2025)	55	15/10/2025	09:59	20.9023 N, 54.0024 E	39.3	988	21.4
Trip 5 (2025)	56	15/10/2025	10:38	20.8216 N, 54.2239 E	42.8	988	19.8
Trip 5 (2025)	57	15/10/2025	11:18	20.7350 N, 54.4176 E	39.4	986	20.3
Trip 5 (2025)	58	15/10/2025	11:41	20.6654 N, 54.5933 E	41.8	986	18.3
Trip 5 (2025)	59	15/10/2025	11:59	20.6563 N, 54.6359 E	40.2	984	20.7
Trip 5 (2025)	60	15/10/2025	13:55	20.7064 N, 54.7321 E	35.9	988	25.5
Trip 5 (2025)	oil spill	12/10/2025	10:20	19.4054 N, 45.7271 E	34.1	924	17.1
Trip 5 (2025)	hot spring	14/10/2025	07:58	20.3972 N, 50.9655 E	32.7	984	22.8

Metadata QC. For Trip 2, the eight values 34.5–41.9 occur in the source humidity column while the temperature column is empty. The ledger assigns them to temperature; neither pressure nor relative humidity is inferred. Trip 3 site 21 is retained as 31.321% RH because

the original workbook records the fraction 0.31321. The 15 March records appended to the legacy Trip 3 worksheet are dated 2023 in that workbook; the curated table therefore labels them Trip 1 auxiliary/revisit records, and Trip 3 ends on 21 February 2024. Trip 5 site 40 contains the source value 194% RH, which is outside the physical range. Because no primary source currently establishes a decimal correction, the curated value is missing; 194 is preserved in the correction ledger.

Table 3: Audited XRF reporting-channel cross-references. Labels, formulas, charge and entity types were checked against pinned ChEBI and PubChem records.

Analyte	Formula/type	ChEBI	PubChem
Ag	Ag	CHEBI:30512	CID:23954
Al	Al	CHEBI:28984	CID:5359268
Al2O3	Al2O3	CHEBI:30187	CID:9989226
As	As	CHEBI:27563	CID:5359596
Au	Au	CHEBI:29287	CID:23985
Ba	Ba	CHEBI:32594	CID:5355457
BaO	BaO	–	CID:62392
Bi	Bi	CHEBI:33301	CID:5359367
Br	Br	CHEBI:22927	CID:5360770
Ca	Ca	CHEBI:22984	CID:5460341
CaO	CaO	CHEBI:31344	CID:14778
Cd	Cd	CHEBI:22977	CID:23973
Ce	Ce	CHEBI:33369	CID:23974
CeO2	CeO2	CHEBI:79089	CID:73963
Cl	Cl	CHEBI:23116	CID:5360523
Co	Co	CHEBI:27638	CID:104730
CoO	CoO	–	CID:14786
Cr	Cr	CHEBI:28073	CID:23976
Cr2O3	Cr2O3	CHEBI:48242	CID:517277
Cs	Cs	CHEBI:30514	CID:5354618
Cu	Cu	CHEBI:28694	CID:23978
CuO	CuO	CHEBI:75955	CID:164827
Dy	Dy	CHEBI:33377	CID:23912
Er	Er	CHEBI:33379	CID:23980
Eu	Eu	CHEBI:32999	CID:23981
Fe	Fe	CHEBI:18248	CID:23925

Analyte	Formula/type	ChEBI	PubChem
Fe2O3	Fe2O3	CHEBI:50819	CID:518696
Ga	Ga	CHEBI:49631	CID:5360835
Ga2O3	Ga2O3	–	CID:158605
Gd	Gd	CHEBI:33375	CID:23982
Ge	Ge	CHEBI:30441	CID:6326954
Hf	Hf	CHEBI:33343	CID:23986
Hg	Hg	CHEBI:25195	CID:23931
Ho	Ho	CHEBI:49648	CID:23988
I	I	CHEBI:24859	CID:5360629
Ir	Ir	CHEBI:49666	CID:23924
K	K	CHEBI:26216	CID:5462222
K2O	K2O	CHEBI:88321	CID:9989219
LE	instrument_pseudoanalyte	–	–
La	La	CHEBI:33336	CID:23926
La2O3	La2O3	–	CID:150906
Lu	Lu	CHEBI:33382	CID:23929
Mg	Mg	CHEBI:25107	CID:5462224
MgO	MgO	CHEBI:31794	CID:14792
Mn	Mn	CHEBI:18291	CID:23930
MnO	MnO	–	CID:14940
Mo	Mo	CHEBI:28685	CID:23932
Na	Na	CHEBI:26708	CID:5360545
Na2O	Na2O	–	CID:73971
Nb	Nb	CHEBI:33344	CID:23936
Nd	Nd	CHEBI:33372	CID:23934
Nd2O3	Nd2O3	–	CID:159373
Ni	Ni	CHEBI:28112	CID:935
NiO	NiO	–	CID:14805

Analyte	Formula/type	ChEBI	PubChem
Os	Os	CHEBI:30687	CID:23937
P	P	CHEBI:28659	CID:5462309
P2O5	P2O5	CHEBI:37376	CID:6326975
Pb	Pb	CHEBI:25016	CID:5352425
PbO	PbO	CHEBI:81045	CID:14827
Pr	Pr	CHEBI:49828	CID:23942
PrO2	PrO2	—	—
Pt	Pt	CHEBI:33364	CID:23939
Rb	Rb	CHEBI:33322	CID:5357696
Re	Re	CHEBI:49882	CID:23947
S	S	CHEBI:26833	CID:5362487
SO3	SO3	CHEBI:29384	CID:24682
Sb	Sb	CHEBI:30513	CID:5354495
Sc	Sc	CHEBI:33330	CID:23952
Sc2O3	Sc2O3	—	CID:134661
Se	Se	CHEBI:27568	CID:6326970
Si	Si	CHEBI:27573	CID:5461123
SiO2	SiO2	CHEBI:30563	CID:24261
Sm	Sm	CHEBI:33374	CID:23951
Sm2O3	Sm2O3	—	CID:159425
Sn	Sn	CHEBI:27007	CID:5352426
Sr	Sr	CHEBI:33324	CID:5359327
SrO	SrO	—	CID:73975
Ta	Ta	CHEBI:33348	CID:23956
Tb	Tb	CHEBI:33376	CID:23958
Th	Th	CHEBI:33385	CID:23960
Ti	Ti	CHEBI:33341	CID:23963
TiO2	TiO2	CHEBI:32234	CID:26042

Analyte	Formula/type	ChEBI	PubChem
Tm	Tm	CHEBI:33380	CID:23961
U	U	CHEBI:27214	CID:23989
V	V	CHEBI:27698	CID:23990
V2O5	V2O5	CHEBI:30045	CID:14814
W	W	CHEBI:27998	CID:23964
Y	Y	CHEBI:33331	CID:23993
Yb	Yb	CHEBI:33381	CID:23992
Zn	Zn	CHEBI:27363	CID:23994
ZnO	ZnO	CHEBI:36560	CID:3007857
Zr	Zr	CHEBI:33342	CID:23995
ZrO2	ZrO2	–	CID:62395

Interpretation. A dash denotes an audited null, not a missing review. These `rdfs:seeAlso` cross-references identify the element or oxide-equivalent reporting channel; they do not assert that the corresponding molecule or mineral occurs in a sample. The P_2O_5 channel is cross-referenced to the ChEBI P_4O_{10} class only because the formulas share the P:O empirical ratio. LE is an instrument aggregate, and PrO_2 has no formula-matching external identifier in the audited sources.

References

- [1] Degtyarenko, K., Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcántara, R., Darsow, M., Guedj, M., Ashburner, M.: ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36**(suppl_1), 453–462 (2008)
- [2] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., *et al.*: PubChem 2023 update. *Nucleic Acids Research* **51**(D1), 1373–1380 (2023)